

Helpdesk & System Troubleshooting Report

Comprehensive diagnostic steps for non-powering laptops, blank monitors, wireless input isolation, and gaming cafe thermal management.

Question 1

A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Non-Powering Laptop Diagnostic Steps:

- **Step 1 (Wall Outlet & Power Source Check):** Plug another device or lamp into the same wall outlet, or try a different socket, to verify whether AC power is active and the outlet itself isn't tripped or faulty.
- **Step 2 (AC Adapter & Cable Inspection):** Test the power brick and cable continuity. Check for fraying, loose barrel/USB-C connectors, or indicator LEDs on the adapter to ensure it is actively outputting DC voltage.
- **Step 3 (Hard Reset / Power Drain):** Disconnect the charger and all USB peripherals, then press and hold the laptop power button continuously for 20 to 30 seconds to drain residual static charge from the motherboard capacitors.
- **Step 4 (Battery Isolation Test):** Remove the external battery (or internally disconnect the battery ribbon cable on modern laptops) and attempt to power on using *only* the direct AC wall adapter to see if a shorted battery is preventing boot.
- **Step 5 (Internal Component & Motherboard Check):** Inspect the DC-in charging port for physical damage, bent pins, or lint obstruction, and check internal rails for short circuits or blown fuses.

Question 2

You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Blank Screen Troubleshooting Flow:

- **Step 1 (Physical Display Connections):** Check that the video cable (HDMI, DisplayPort, VGA) is securely plugged into both the monitor and the back of the PC. If using a discrete graphics card, ensure the cable is plugged into the GPU ports, not the motherboard.
- **Step 2 (Monitor Power & Input Source):** Verify that the monitor's power cable is firmly connected and the power LED is on. Press the monitor's physical menu buttons to manually cycle through input sources (e.g., HDMI 1, DisplayPort) to ensure it matches the active port.
- **Step 3 (Monitor Self-Test):** Disconnect the video cable entirely from the PC; a healthy monitor should display an on-screen "No Signal" floating dialog box, ruling out a dead display panel.
- **Step 4 (Hardware Reseating):** Power down the system completely, open the chassis, and reseat the graphics card and RAM modules, as loose seating frequently cuts off video output signals during POST.
- **Step 5 (Display Settings Reset):** Boot the system into Windows Safe Mode or use another monitor to check if an incorrect resolution or multi-display setting is causing the blank screen.

Question 3

A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

Wireless Mouse Isolation Procedure:

- **Step 1 (Power & Battery Check):** Check the physical power switch underneath the mouse and replace the AA/AAA batteries or place the rechargeable mouse on its charging dock to verify it has sufficient power.
- **Step 2 (USB Receiver Port Isolation):** Unplug the wireless USB dongle from its current port and plug it into a known working USB port (such as the port currently working for the keyboard) to determine if the original USB port is faulty.
- **Step 3 (Cross-System Testing):** Plug the wireless USB receiver into a different computer or laptop. If the mouse works there, the mouse hardware is fine and the issue lies within the original system's software or drivers.
- **Step 4 (Device Manager & Driver Verification):** Open Windows Device Manager (`devmgmt.msc`), check under "Mice and other pointing devices" or "Universal Serial Bus controllers" for yellow warning triangles, uninstall corrupted HID drivers, and allow Windows to reinstall them automatically.

Question 4

Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Gaming Cafe Overheating Causes & Inspection Methods:

- **Cause 1 (Dust Accumulation & Blocked Airflows):** Gaming cafe environments accumulate heavy dust quickly. *Check:* Open the chassis side panel and inspect intake filters, heatsink fins, and fan blades. Clean thoroughly using compressed air.
- **Cause 2 (CPU Cooler Fan Failure / Poor Mounting):** If a fan is seized, spinning too slowly, or the CPU cooler bracket is loose. *Check:* Power on the system and visually observe if all chassis, GPU, and CPU cooler fans are spinning properly while monitoring idle/load temperatures using software like HWMonitor.
- **Cause 3 (Degraded or Dried-Out Thermal Paste):** Thermal paste between the CPU IHS and heatsink can dry out over time under intense gaming loads. *Check:* Remove the CPU cooler heatsink, inspect the thermal paste spread and texture; if dry, flaky, or cracked, clean the surface with isopropyl alcohol and apply a fresh layer of high-performance thermal compound.